BSCE Board Exam Practice Questions

Question 1 (Math)

A steel rod 10 meters long has a cross-sectional area of 0.002 m^2. If a tensile force of 50,000 N is applied, what is the stress in the rod?

A. 20,000,000 Pa
B. 25,000,000 Pa
C. 30,000,000 Pa
D. 35,000,000 Pa
E. 40,000,000 Pa

Correct Answer: B

Question 2 (Mathematic)

A simply supported beam of 6 meters span carries a uniformly distributed load of 12 kN/m. What is the maximum bending moment at midspan?

A. 48 kN·m
B. 54 kN·m
C. 60 kN·m
D. 72 kN·m
E. 108 kN·m

Correct Answer: 54 kN·m

Question 3 (Mathematics)

A concrete mix has a water-cement ratio of 0.45. If the weight of cement used is 400 kg, what is the weight of water required?

A. 160 kg
B. 175 kg
C. 180 kg
D. 190 kg
E. 200 kg

Correct Answer: C

Question 4 (Mathematics)

A circular column has a diameter of 500 mm and is subjected to an axial compressive load of 1,500 kN. What is the average compressive stress?

A. 5.86 MPa
B. 6.43 MPa
C. 7.07 MPa
D. 7.64 MPa
E. 8.12 MPa

Correct Answer: D

Question 5

Which of the following materials has the highest compressive strength?

A. Timber

B. Brick masonry
C. Reinforced concrete
D. Structural steel
E. Fiber reinforced polymer

Question 6
Refer to Image 1. Based on the truss diagram shown, what is the force in member BC if the applied load at joint B is 80 kN downward?

Image 1: A simple Pratt truss with 3 panels, pinned at A, roller at D, with an 80 kN downward load at joint B

A. 40 kN tension
B. 60 kN compression
C. 80 kN tension
D. 80 kN compression
E. 100 kN tension
Correct Answer: C

Question 7
Refer to Image 2. Based on the shear force diagram shown, at what location along the beam does the maximum bending moment occur?

A. At the left support
B. At 1 meter from the left support
C. At midspan
D. At 2 meters from the right support
E. At the right support
Correct Answer: C
(Image 2: A shear force diagram of a simply supported beam with a point load at midspan, showing positive shear on the left half and negative shear on the right half)